ACTL2102 Tutorial

Week 8: Occupancy, Limiting Probabilities and the Embedded Chain

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T3 2026

Plan

- Recap: time in a state, the long run, the embedded chain
- Main questions: **4.17, 4.3, 4.12, 4.13, 4.14 (parts 1–5)**

Notation from Week 7

$P_{ij}(t)$	probability of being in j after time t, from i, by any route
ν_i	rate of leaving i; holding time $T_i \sim \text{Exp}(\nu_i)$, mean $1/\nu_i$
P_{ij}	probability the next state is j, given the chain leaves i
q_{ij}	rate of moves $i \to j$: $\nu_i P_{ij}$, the derivative of $P_{ij}(t)$ at $t = 0$
Q	generator: q_{ij} off the diagonal, $-\nu_i$ on it; rows sum to 0
$o(h)$	negligible next to h: $o(h)/h \to 0$ as $h \to 0$

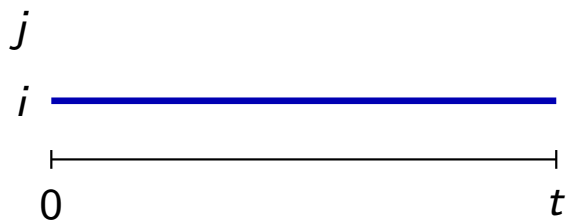
Weeks 7 and 8 are homogeneous: rates are constant in time. The Kolmogorov equations are a system of ODEs in the $P_{ij}(t)$, solved together from $P(0) = I$.

Occupancy Probabilities

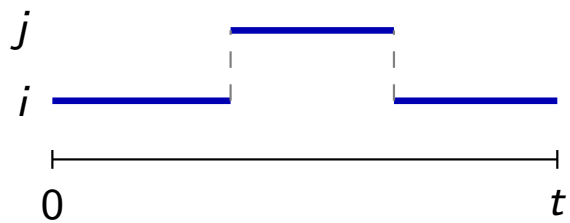
In a homogeneous chain, the **occupancy probability** of staying in i throughout $[0, t]$ is

$$\overline{P}_{ii}(t) = \Pr(T_i > t) = e^{-\nu_i t}.$$

The lecture puts the bar over the subscript, $P_{\overline{ii}}(t)$. The bar separates it from the transition probability $P_{ii}(t)$, which also counts paths that leave i and return.



counted by $\overline{P}_{ii}(t)$ and $P_{ii}(t)$



counted by $P_{ii}(t)$ only

By memorylessness, the remaining time in i is $\text{Exp}(\nu_i)$, however long the chain has been there.

Exercise 4.17: Retirement home

Exercise 4.17: [cmc16, Solution] A retirement home has separate accommodation for healthy residents and a nursing facility for ill residents. Residents who get ill may spend time in the nursing facility on a temporary basis if they recover. Residents may leave the retirement home because of death or because they move elsewhere. Consider a continuous time Markov chain model of the retirement home residents with the following states:

State	
1	Healthy - Separate Accommodation
2	Ill - Nursing Facility
3	Dead
4	Withdrawn from Retirement Home

Transition rates between the states are denoted by σ_{ij}.
Assume the following transition rates per year apply:

$$\sigma_{12} = 0.4, \quad \sigma_{13} = 0.1, \quad \sigma_{14} = 0.2,$$
$$\sigma_{21} = 0.5, \quad \sigma_{23} = 0.4, \quad \sigma_{24} = 0.3,$$

and for all other $i \neq j$ we have $\sigma_{ij} = 0$.
Justifying the formula that you use, determine:

1. the probability that a healthy resident will remain healthy for at least 5 years;
2. the probability that a healthy resident will withdraw from the retirement home before getting ill over the next year;
3. the probability that a resident entering the nursing home will recover within 2 years;
4. the probability that a healthy resident will become ill, enter the nursing home and remain in the nursing home until the end of the year.

Limiting Probabilities

The **limiting probabilities**, when they exist and do not depend on the starting state i, are

$$P_j = \lim_{t \to \infty} P_{ij}(t).$$

As in discrete time, they exist when the chain is **ergodic**: all states communicate, and are positive recurrent and aperiodic. In continuous time every state is automatically aperiodic, as $P_{ii}(t) > 0$ for all t.

Classes, recurrence and transience are as in discrete time, with an arrow from i to j wherever $q_{ij} > 0$. In a finite chain, closed classes are positive recurrent and the other states transient, so the chain is ergodic if all states communicate.

Discrete and Continuous Time

In an ergodic chain, discrete or continuous, the limit of the transition probabilities, the stationary probabilities and the long-run proportions of time are the same numbers.

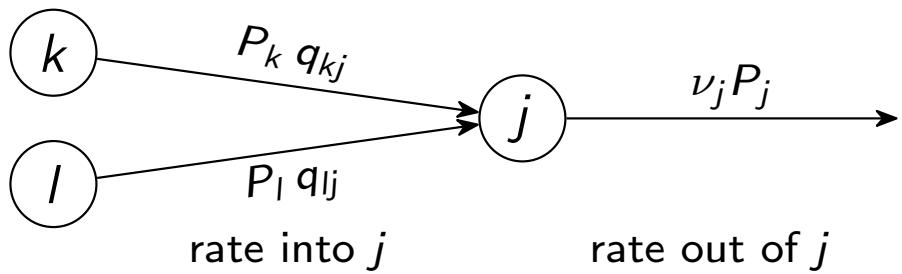
	Discrete (Week 3) $\boldsymbol{\pi} = (\pi_1, \pi_2, \dots)$	Continuous $\boldsymbol{P} = (P_1, P_2, \dots)$
Limit	$\pi_j = \lim_{n\to\infty} P_{ij}^{(n)}$	$P_j = \lim_{t\to\infty} P_{ij}(t)$
Stationary	$X_0 \sim \boldsymbol{\pi} \Rightarrow X_n \sim \boldsymbol{\pi}$	$X(0) \sim \boldsymbol{P} \Rightarrow X(t) \sim \boldsymbol{P}$
Long-run proportion	π_j of the time in j	P_j of the time in j
Ergodic	positive recurrent and aperiodic	positive recurrent, aperiodic automatically

A periodic discrete chain has stationary probabilities but no limit. This cannot happen in continuous time.

Balance Equations

The limiting probabilities solve the **balance equations**:

$$\nu_j P_j = \sum_{k \neq j} P_k \, q_{kj} \quad \text{for every } j, \qquad \sum_k P_k = 1.$$



Equivalently, $\boldsymbol{P}Q = \mathbf{0}$, where the row vector $\boldsymbol{P} = (P_1, P_2, \dots)$ is the counterpart of π in discrete time. The equation for P_j is read down column j of Q. One equation is redundant, and $\sum_k P_k = 1$ replaces it.

Exercise 4.3: Bermudan option: matching discrete and continuous limiting probabilities

Exercise 4.3: [cmc2, Solution] The probability that a Bermudan option[1] is in-the-money on the exercise date is modelled using a Markov chain. The exercise dates of this option are the first trading day of the year and the probability that the option is in-the-money given that it was last year in-the-money is 0.6 and the probability that the option is in-the-money given that it was last year out-the-money is 0.2. The insurance company wants to adjust their modelling such that it can calculate the probability of the option being in-the-money the next exercise date at any time in the year.

1. Find the transition probability matrix and the limiting probabilities.
2. Denote α is the rate out of option is in-the-money and β is the rate out of option is out-the-money. Construct the transition rates matrix of the continuous model. By equating the the limiting probabilities in the continuous time and discrete time, show that:

$$\alpha = 2\beta.$$

[1] Bermudan option contracts are option which can be exercised on specified dates.

Exercise 4.12: Two generators: holding times, classes, limiting probabilities

Exercise 4.12: [cmc11, Solution] For the following infinitesimal generators (i.e., matrices with transition rates):

$$\mathbf{Q}_1 = \begin{bmatrix} -0.4 & 0.3 & 0 & 0.1 & 0 \\ 0 & -0.8 & 0.6 & 0.2 & 0 \\ 0.3 & 0 & -0.9 & 0.6 & 0 \\ 0 & 0 & 0 & -0.3 & 0.3 \\ 0 & 0 & 0 & 0.2 & -0.2 \end{bmatrix} \quad \mathbf{Q}_2 = \begin{bmatrix} -0.4 & 0.3 & 0 & 0.1 \\ 0 & -0.8 & 0.6 & 0.2 \\ 0.3 & 0 & -0.9 & 0.6 \\ 0.3 & 0 & 0 & -0.3 \end{bmatrix}.$$

1. What is the expected time spent in each state before a transition to another state, conditional on being in that state.
2. What is the probability that the time spent in the last state of each process is smaller than one year?
3. Classify the classes of states. Are the states recurrent or transient.
4. Which processes are ergodic?
5. Find the limiting probabilities.

The Embedded Chain

Ignoring time, the sequence of states visited is a discrete-time Markov chain, the **embedded chain**, with transition matrix $\mathbf{P}_{\text{emb}}$ of jump probabilities P_{ij}. Its stationary probabilities solve the same equations as in Week 3:

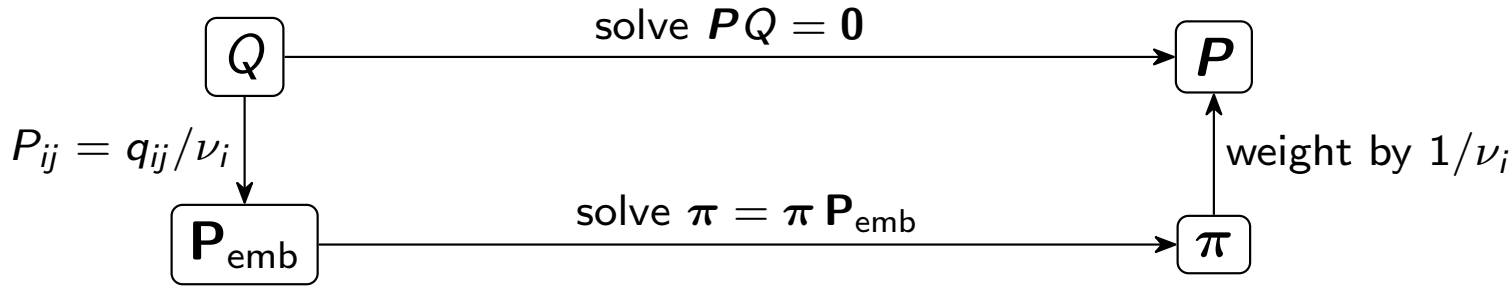
$$\boldsymbol{\pi} = \boldsymbol{\pi}\,\mathbf{P}_{\text{emb}}, \qquad \sum_k \pi_k = 1.$$

π_i	long-run proportion of *transitions* into i
P_i	long-run proportion of *time* in i

These differ, as π_i ignores the length of visits, and agree only when all the ν_i are equal.

Example. The chain A, B, C from Week 7 has $\boldsymbol{\pi} = (\frac{1}{3}, \frac{4}{9}, \frac{2}{9})$ and $\boldsymbol{P} = (\frac{2}{5}, \frac{2}{5}, \frac{1}{5})$. State B receives the most transitions, but $\nu = (3, 4, 4)$ makes its visits shorter than those to A.

Two Routes to the Limiting Probabilities



The P_i come directly from Q, or from the embedded chain by a formula connecting π_i to P_i:

$$P_i = \frac{\pi_i/\nu_i}{\sum_k \pi_k/\nu_k}.$$

Weighting by the mean visit length $1/\nu_i$ turns proportions of transitions into proportions of time.

Example. For the chain A, B, C, $\pi_i/\nu_i = (\frac{1}{9}, \frac{1}{9}, \frac{1}{18})$ normalises to $(\frac{2}{5}, \frac{2}{5}, \frac{1}{5})$, as from the balance equations.

Exercise 4.13: Limits through the embedded chain

Exercise 4.13: [cmc12, Solution] For the following transition intensity matrix:

$$\mathbf{Q} = \begin{bmatrix} -0.9 & 0.2 & 0.3 & 0.4 \\ 0.2 & -1.6 & 0.7 & 0.7 \\ 0.1 & 0.4 & -1.3 & 0.8 \\ 0.3 & 0.2 & 1 & -1.5 \end{bmatrix}.$$

1. Give the limiting probabilities.
2. Give the transition probabilities of the process of the sequences of states visited.
3. Give the limiting proportion of the transitions into each state.
4. Use the result in (3) to check the limiting probabilities of being in a state (in the original Markov chain) you obtained in (1).

Exercise 4.14: Stock market states (parts 1–5)

Exercise 4.14: [cmc13, Solution] The return dynamics of a stock depends on the whether the market is booming, bursting, or in a normal state. Moreover, if the company is bankrupt the return dynamics are different. It is given that the yearly transition rates are:

- from normal state to booming state is 1/3;
- from normal state to busting state is 1/4;
- from normal state to bankrupt state is 1/40;
- from busting state to bankrupt state is 1/10;
- from booming state to bankrupt state is 1/100;
- from busting state to normal state is 1/6;
- from booming state to normal state is 1/2.

1. Give the infinitesimal generator (i.e., the matrix with transition rates).
2. Calculate the expected remaining time in the booming state before going to the normal or bankrupt state conditional on being in the state for two years. Also calculate the expected total time in the bursting state before going to the normal or bankrupt state conditional on being in the state for one year.
3. What is the probability that the bursting period will take five years or longer?
4. Consider the process of the states visited over time. Give the transition matrix of this process. Note: Assume that the next state visited for the bankrupt state is the bankrupt state.
5. Give the transition probability matrix of the original process between time t and time t.

Notation from This Week

$\overline{P}_{ii}(t)$	occupancy: stays in i throughout $[0, t]$; equals $e^{-\nu_i t}$
P_j	limit of $P_{ij}(t)$, stationary probability and long-run proportion of time in j
$\boldsymbol{P}$	row vector $(P_1, P_2, \dots)$ solving $\boldsymbol{P}Q = \mathbf{0}$; counterpart of $\boldsymbol{\pi}$ in discrete time
$\mathbf{P}_{\text{emb}}$	transition matrix of the embedded chain, entries P_{ij}
π_j	embedded chain: long-run proportion of transitions into j
Link	P_i is proportional to π_i/ν_i

Homogeneous and Non-Homogeneous Chains

	Homogeneous (Weeks 7–8)	Non-homogeneous (Week 9)
Rates	q_{ij}, constant	$q_{ij}(t)$, vary with time
Transition probabilities	$P_{ij}(t)$: elapsed time	$P_{ij}(s,t)$: both times
Holding times	$T_i \sim \text{Exp}(\nu_i)$	not exponential in general
Occupancy	$e^{-\nu_i t}$	$\exp\left(-\int_s^t \nu_i(u)\,du\right)$
Kolmogorov equations	constant coefficients	time-varying coefficients; numerical methods in general

Rates that depend on age, as in most actuarial models, make a chain non-homogeneous.

Thanks for coming!

Appendix: Setting Up a First-Jump Integral

Let $X(0) = i$, let $T_i \sim \text{Exp}(\nu_i)$ be the time of the first jump, and let $J = X(T_i)$ be the state entered, with $\Pr(J = j) = P_{ij}$ independently of T_i. For $A = \{T_i \le t,\ J = j\}$, $\Pr(A \mid T_i = u)$ is P_{ij} if $u \le t$ and 0 otherwise, so by the law of total probability

$$\Pr(A) = \int_0^t \underbrace{\nu_i e^{-\nu_i u}}_{\text{pdf of } T_i} P_{ij}\, du = \int_0^t \underbrace{\overline{P}_{ii}(u)}_{\text{stays until } u} \underbrace{q_{ij}\, du}_{\text{moves to } j},$$

where $\overline{P}_{ii}(u) = e^{-\nu_i u}$ is the survival function of T_i and $q_{ij} = \nu_i P_{ij}$. For $B = A \cap \{X(s) = j \text{ for } T_i \le s \le t\}$, the Markov property at T_i adds a factor $\overline{P}_{jj}(t-u)$ inside the integral.

